

Hugo Chen

+1 236-412-0876 | hugohqchen@gmail.com | github.com/Bobwillrule | Personal Site

WORK EXPERIENCE

Software Engineer Intern

Jul. 2026 – Sept. 2026

TSMC (Taiwan Semiconductor Manufacturing Company)

Hsinchu, Taiwan

- Built an AI assistant that analyzes logs from 60,000+ semiconductor machines and correlates failures with relevant codebases, accelerating root-cause analysis and internal debugging
- Collaborated with 9 other interns using Git to engineer an OpenHarness-based multi-agent system for LLM orchestration with custom MCP servers, tools, skills, and retrieval pipelines
- Reduced end-to-end agent execution time by 40% by developing a full-stack platform with React, FastAPI, Python, and MongoDB to replace OpenHarness's CLI-based process and agent loop
- Implemented Azure DevOps CI/CD pipelines to deploy containerized AI services using Docker and Kubernetes to TSMC's internal testing environment for department-wide use across 200 engineers

Web Developer & Carpenter

Jun. 2023 – Oct. 2024

Linty Constructions

Vancouver, BC

- Generated 3 new client leads in 2 months by creating company website that increased online visibility
- Framed 4 custom residential homes while adapting to fast-paced outdoor job site conditions
- Acted as the English-Mandarin liaison between clients and contractors, reducing miscommunication

Bike Mechanic

Jun. 2022 – Jun. 2023

Canadian Tire

Coquitlam, BC

- Maintained the store's lowest bike rework and return rates through systematic inspection and tuning
- Advised customers on bike selection and achieved the store's top monthly bike sales

PROJECTS

AI Trader | *Python, PyTorch, Pandas, yFinance*

Aug. 2025 – Present

- Built a deep reinforcement-learning system in Python and PyTorch, training a Deep Q-Network (DQN) on 100K+ historical market observations using chronological train, validation, and test split
- Created a trading environment with fees, slippage, and model-selected stop-loss/take-profit strategies
- Shipped a FastAPI dashboard for training orchestration and policy benchmarking against cash and buy-and-hold, with Sharpe ratio and maximum-drawdown metrics

BehindTheETF | *Next.js, React, TypeScript, PostgreSQL*

Jun. 2026 – Present

- Built an ETF analytics platform with recursive fund look-through beyond standard portfolio tools
- Designed a TypeScript ingestion pipeline for 5 ETF providers and issuer-specific data integrations
- Implemented APIs for holdings, sector, country, and overlap exposure across nested ETFs

Financial Tracker | *Java, Swing*

Sep. 2025 – Jan 2026

- Created a Java Swing desktop app to track income, expenses, and account balances
- Designed an object-oriented architecture for managing transactions, balances, and account state
- Wrote JUnit tests with 100% coverage to validate core financial logic and improve model correctness

EDUCATION

University of British Columbia

Vancouver, Canada

B.A.Sc. in Electrical Engineering, Minor in Computer Science

2025 – 2029

- GPA: 89%, Dean's List
- Relevant coursework: Probability Theory, Data Structures and Algorithms I & II, Machine Learning, Computing Systems, Signals & Systems, Circuit Analysis, Differential Equations, Linear Algebra, Calc 3

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C/C++, Kotlin, HTML/CSS, MATLAB, Assembly

Frameworks/Libraries: React, Pandas, NumPy, Scikit-learn, PyTorch, FastAPI

Tools/Platforms: OpenAI Codex, Opencode, Docker, Kubernetes, Azure DevOps, Git, WordPress, AutoCAD

Spoken Languages: English (Native), Mandarin (Native), Japanese (Advanced)